

LAB SHEET



K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B. TECH CSE (CORE)

Database & Management System Lab (ENCS254)

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LAB SHEET - 1

Objective: To design ER diagrams and convert entities and relationships into relational tables using proper schema notation.

• DDL Commands

1. Create Database

- `CREATE DATABASE LAB_SHEET_1;`
- `USE LAB_SHEET_1;`

• PART A – COLLEGE DATABASE

1. Create Tables with Primary and Foreign Keys

```
• CREATE TABLE STUDENT (  
    USN VARCHAR(10) PRIMARY KEY,  
    SName VARCHAR(50),  
    Address VARCHAR(100),  
    Phone VARCHAR(15),  
    Gender VARCHAR(10)  
);
```

2. Create SEMSEC Table

```
• CREATE TABLE SEMSEC (  
    SSID INT PRIMARY KEY,  
    Sem INT,  
    Sec VARCHAR(5)  
);
```

3. Create CLASS Table

```
• CREATE TABLE CLASS (  
    USN VARCHAR(10),  
    SSID INT,  
    PRIMARY KEY (USN, SSID),  
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),  
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)  
);
```

4. Create SUBJECT Table

```

CREATE TABLE SUBJECT (
    Subcode VARCHAR(10) PRIMARY KEY,
    Title VARCHAR(50),
    Sem INT,
    Credits INT
);

```

5. Create IAMARKS Table

```

CREATE TABLE IAMARKS (
    USN VARCHAR(10),
    Subcode VARCHAR(10),
    SSID INT,
    Test1 INT,
    Test2 INT,
    Test3 INT,
    FinalIA INT,
    PRIMARY KEY (USN, Subcode, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (Subcode) REFERENCES SUBJECT(Subcode),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);

```

PART B – COMPANY DATABASE

1. Create DEPARTMENT Table

```

CREATE TABLE DEPARTMENT (
    Dno INT PRIMARY KEY,
    Dname VARCHAR(50),
    Mgrssn VARCHAR(15),
    Mgrstartdate DATE
);

```

2. Create EMPLOYEE Table

```

CREATE TABLE EMPLOYEE (
    SSN VARCHAR(15) PRIMARY KEY,
    Name VARCHAR(50),
    Address VARCHAR(100),
    Sex VARCHAR(10),
    Salary DECIMAL(10,2) NOT NULL,
    Superssn VARCHAR(15),
    Dno INT,
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);

```

3. Create DLocation Table

```

CREATE TABLE DLOCATION (
    Dno INT,
    Dloc VARCHAR(50),
    PRIMARY KEY (Dno, Dloc),
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);

```

4. Create PROJECT Table

```

CREATE TABLE PROJECT (
    PNo INT PRIMARY KEY,
    PName VARCHAR(50),
    PLocation VARCHAR(50),
    DNo INT,
    FOREIGN KEY (DNo) REFERENCES DEPARTMENT(Dno)
);

```

5. Create WORKS_ON Table

```

CREATE TABLE WORKS_ON (
    SSN VARCHAR(15),
    PNo INT,
    Hours DECIMAL(5,2),
    PRIMARY KEY (SSN, PNo),
    FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN),
    FOREIGN KEY (PNo) REFERENCES PROJECT(PNo)
);

```

• PART C – CUSTOMER SALE DATABASE

1. Create CUSTOMER Table

```

CREATE TABLE CUSTOMER (
    Cust_id INT PRIMARY KEY,
    Cust_name VARCHAR(50)
);

```

2. Create ITEM Table

- ```
CREATE TABLE ITEM (
 Item_id INT PRIMARY KEY,
 Item_name VARCHAR(50),
 Price DECIMAL(10,2) NOT NULL
);
```

### 3. Create SALE Table

- ```
CREATE TABLE SALE (  
    Bill_no INT PRIMARY KEY,  
    Bill_date DATE,  
    Cust_id INT,  
    Item_id INT,  
    Qty_sold INT,  
    FOREIGN KEY (Cust_id) REFERENCES CUSTOMER(Cust_id),  
    FOREIGN KEY (Item_id) REFERENCES ITEM(Item_id)  
);
```

• ALTER OPERATIONS

1. Add Email Column

- ```
ALTER TABLE STUDENT
ADD Email VARCHAR(100);
```

### 2. Rename SName to StudentName

- ```
ALTER TABLE STUDENT  
CHANGE SName StudentName VARCHAR(50);
```

3. Drop DLOCATION Table

- ```
DROP TABLE DLOCATION;
```

## • DML Commands

1. Inserted 10 records into CUSTOMER, ITEM and SALE tables.

```

• INSERT INTO CUSTOMER VALUES (1,'Rohan');
• INSERT INTO CUSTOMER VALUES (2,'Priya');
• INSERT INTO CUSTOMER VALUES (3,'Aman');
• INSERT INTO CUSTOMER VALUES (4,'Neha');
• INSERT INTO CUSTOMER VALUES (5,'Karan');
• INSERT INTO CUSTOMER VALUES (6,'Simran');
• INSERT INTO CUSTOMER VALUES (7,'Arjun');
• INSERT INTO CUSTOMER VALUES (8,'Meera');
• INSERT INTO CUSTOMER VALUES (9,'Rahul');
• INSERT INTO CUSTOMER VALUES (10,'Anjali');

• INSERT INTO ITEM VALUES (101,'Laptop',50000);
• INSERT INTO ITEM VALUES (102,'Mouse',500);
• INSERT INTO ITEM VALUES (103,'Keyboard',1500);
• INSERT INTO ITEM VALUES (104,'Bag',1200);
• INSERT INTO ITEM VALUES (105,'Shoes',2500);
• INSERT INTO ITEM VALUES (106,'Watch',3000);
• INSERT INTO ITEM VALUES (107,'Book',300);
• INSERT INTO ITEM VALUES (108,'Pen',50);
• INSERT INTO ITEM VALUES (109,'Mobile',20000);
• INSERT INTO ITEM VALUES (110,'Headphones',1800);

• INSERT INTO SALE VALUES (1001,CURDATE(),1,101,1);
• INSERT INTO SALE VALUES (1002,CURDATE(),2,102,2);
• INSERT INTO SALE VALUES (1003,CURDATE(),3,105,1);
• INSERT INTO SALE VALUES (1004,CURDATE(),4,109,1);
• INSERT INTO SALE VALUES (1005,CURDATE(),5,103,3);
• INSERT INTO SALE VALUES (1006,CURDATE(),6,104,2);
• INSERT INTO SALE VALUES (1007,CURDATE(),7,106,1);
• INSERT INTO SALE VALUES (1008,CURDATE(),8,107,5);

• INSERT INTO SALE VALUES (1009,CURDATE(),9,110,2);
• INSERT INTO SALE VALUES (1010,CURDATE(),5,101,1);

```

## 2. Bills of Current Date:

- ```
SELECT S.Bill_no, C.Cust_name, I.Item_name
FROM SALE S
JOIN CUSTOMER C ON S.Cust_id = C.Cust_id
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE S.Bill_date = CURDATE();
```

Output:

Result Grid   Filter Rows:			
	Bill_no	Cust_name	Item_name
▶	1001	Rohan	Laptop
	1002	Priya	Mouse
	1003	Aman	Shoes
	1004	Neha	Mobile
	1005	Karan	Keyboard
	1006	Simran	Bag
	1007	Arjun	Watch
	1008	Meera	Book
	1009	Rahul	Headphones
	1010	Karan	Laptop

3. Total Bill Amount:

- ```
SELECT S.Bill_no, (S.Qty_sold * I.Price) AS Total_Amount
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id;
```

Output:

| Result Grid   Filter Rows: <input type="text"/> |         |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------|
|                                                                                                                                                                                                                       | Bill_no | Total_Amount |
| ▶                                                                                                                                                                                                                     | 1001    | 50000.00     |
|                                                                                                                                                                                                                       | 1002    | 1000.00      |
|                                                                                                                                                                                                                       | 1003    | 2500.00      |
|                                                                                                                                                                                                                       | 1004    | 20000.00     |
|                                                                                                                                                                                                                       | 1005    | 4500.00      |
|                                                                                                                                                                                                                       | 1006    | 2400.00      |
|                                                                                                                                                                                                                       | 1007    | 3000.00      |
|                                                                                                                                                                                                                       | 1008    | 1500.00      |
|                                                                                                                                                                                                                       | 1009    | 3600.00      |
|                                                                                                                                                                                                                       | 1010    | 50000.00     |

4. Customers who purchased products priced above 200.



- ```
SELECT DISTINCT C.Cust_name
FROM SALE S
JOIN CUSTOMER C ON S.Cust_id = C.Cust_id
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE I.Price > 200;
```

Output:

	Cust_name
▶	Rohan
	Karan
	Priya
	Simran
	Aman
	Arjun
	Meera
	Neha
	Rahul

- Count number of products purchased by each customer.

- ```
SELECT C.Cust_name, COUNT(S.Item_id) AS Product_Count
FROM SALE S
JOIN CUSTOMER C ON S.Cust_id = C.Cust_id
GROUP BY C.Cust_name;
```

Output:

| Result Grid     Filter Rows: <input type="text"/> |           |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|
|                                                                                                                                                                                                                         | Cust_name | Product_Count |
| ▶                                                                                                                                                                                                                       | Rohan     | 1             |
|                                                                                                                                                                                                                         | Priya     | 1             |
|                                                                                                                                                                                                                         | Aman      | 1             |
|                                                                                                                                                                                                                         | Neha      | 1             |
|                                                                                                                                                                                                                         | Karan     | 2             |
|                                                                                                                                                                                                                         | Simran    | 1             |
|                                                                                                                                                                                                                         | Arjun     | 1             |
|                                                                                                                                                                                                                         | Meera     | 1             |
|                                                                                                                                                                                                                         | Rahul     | 1             |

- Products purchased by customer with Cust\_id = 5.



```

• SELECT I.Item_name
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE S.Cust_id = 5;

```

Output:

| Item_name |
|-----------|
| Keyboard  |
| Laptop    |

### • Views Created

1. Bill\_Details View (shows bill\_no, bill\_date, cust\_id, item\_id, price, qty\_sold, amount).

```

CREATE VIEW Bill_Details AS
SELECT S.Bill_no, S.Bill_date, S.Cust_id,
 S.Item_id, I.Price, S.Qty_sold,
 (I.Price * S.Qty_sold) AS Amount
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id;

```

```

• SELECT * FROM Bill_Details;

```

Output:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

|   | Bill_no | Bill_date  | Cust_id | Item_id | Price    | Qty_sold | Amount   |
|---|---------|------------|---------|---------|----------|----------|----------|
| ▶ | 1001    | 2026-03-01 | 1       | 101     | 50000.00 | 1        | 50000.00 |
|   | 1002    | 2026-03-01 | 2       | 102     | 500.00   | 2        | 1000.00  |
|   | 1003    | 2026-03-01 | 3       | 105     | 2500.00  | 1        | 2500.00  |
|   | 1004    | 2026-03-01 | 4       | 109     | 20000.00 | 1        | 20000.00 |
|   | 1005    | 2026-03-01 | 5       | 103     | 1500.00  | 3        | 4500.00  |
|   | 1006    | 2026-03-01 | 6       | 104     | 1200.00  | 2        | 2400.00  |
|   | 1007    | 2026-03-01 | 7       | 106     | 3000.00  | 1        | 3000.00  |
|   | 1008    | 2026-03-01 | 8       | 107     | 300.00   | 5        | 1500.00  |
|   | 1009    | 2026-03-01 | 9       | 110     | 1800.00  | 2        | 3600.00  |
|   | 1010    | 2026-03-01 | 5       | 101     | 50000.00 | 1        | 50000.00 |

2. Weekly\_Sales View (shows daily total sales for last one week).

- ```
CREATE VIEW Weekly_Sales AS
SELECT Bill_date, SUM(Qty_sold * Price) AS Total_Sales
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE Bill_date >= CURDATE() - INTERVAL 7 DAY
GROUP BY Bill_date;
```
- ```
SELECT * FROM Weekly_Sales;
```

Output:

| Result Grid   Filter Rows: <input type="text"/> |            |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|
|                                                                                                                                                                                                                   | Bill_date  | Total_Sales |
| ▶                                                                                                                                                                                                                 | 2026-03-01 | 138500.00   |